AI-Powered Personalized OTT Recommendation System

AMachine Learningproject thatdeliverssmart,genre-basedmovie recommendations — built with Python, TF-IDF Vectorization, and Cosine Similarity ontheMovieLens dataset.

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ML PROJECT REPORT

Project Overview

What It Does

Analyzes movie genres and user preferences to surface relevant content — mirroring how Netflix and Amazon Prime drive engagement through intelligent recommendations.

How It Works

Content-basedfiltering compares genre profiles using **TF-IDF Vectorization** to encode text and **Cosine Similarity** to rank movie-to-movie closeness.

Tools, Technologies & Build Process

Python

Core language

Google Colab

Dev environment

Pandas & NumPy

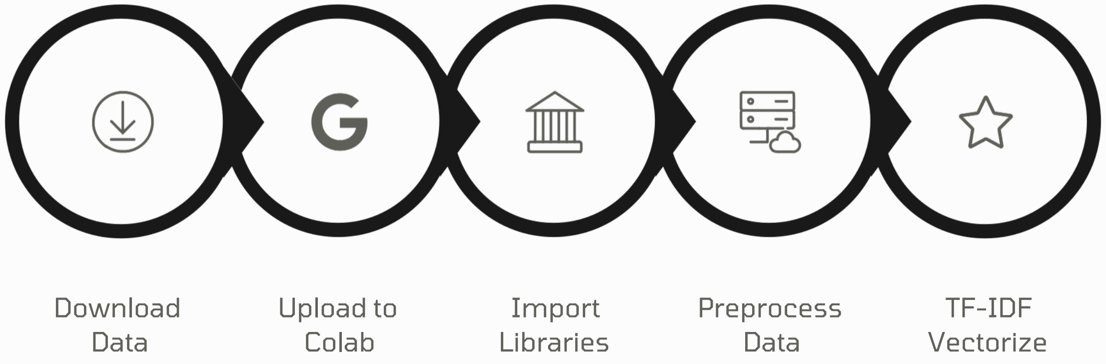
Data handling

Scikit-learn

ML algorithms

MovieLens

Trainingdataset



System Output

Input:**ToyStory(1995)**—thesystemreturns genre-matched recommendations with high cosine similarity scores.

🎬 Jumanji	🎬 Toy Story 2	🎬 Monsters, Inc.
🎬 Finding Nemo	🎬 A Bug's Life	

All recommendations are genre-coherent family/adventure titles — validating the content-based filtering logic.

Conclusions & Future Roadmap

Key Takeaways

The project successfully demonstrates real-world ML concepts — TF-IDF vectorization, cosine similarity, and content-based filtering — in a practical OTT context. It solidifies foundational AI/ML understanding through hands-on implementation.

Deployable as a live web app using **Streamlit** or **Render** with minimal refactoring.

Planned Enhancements

→ **Streamlit Web Interface**
Interactive front-end for live recommendations

→ **TMDB API Integration**
Movie posters and rich metadata display

→ **Mood-Based Filtering**
Emotion-aware recommendation layer

→ **User Auth & Watchlists**
Personalized accounts and saved content